Algorithm Design and Analysis

Course

CCP6214 Algorithm Design and Analysis

Trimester

March/April 2025 (Term 2510)

Meet the Team

Lab Section: T18L

Lab Lecturer: Alandoli Mohammed Nasser Mohammed

Group No.: G2

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Group Members & Contributions

ID	Name	Contribution
1211108820	OW KA SHENG	Dataset Generation, Merge Sort (Java & Python), Quick Sort (Java & Python), Binary Search (Java & Python), Binary Search Step (Java & Python)
1231301318	USUPOV ALIMZHAN	Slide Preparation, Device Screenshot
1221304263	MUHAMMAD HAZIM BIN MUHD SHAFIQ ZULHILMI KEE	Slide Preparation, Device Screenshot, Dataset Upload, Merge Sort Step (Java & Python)
1221307698	GALAL OSMAN GALAL MOHAMED	Slide Preparation, Device Screenshot, Dataset Upload, Quick Sort Step (Java & Python)

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Dataset Generation

Format:

integer,string (e.g., 984324,abcde)

Approach:

- Random, unique positive integers
- String values generated randomly
- Max dataset size: 20 million
- Generated using Python
- File format: .csv

Dataset Links (OneDrive):

- OW KA SHENG
- MUHAMMAD HAZIM BIN MUHD SHAFIQ ZULHILMI KEE
- GALAL OSMAN GALAL MOHAMED
- IUSUPOV ALIMZHAN

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Merge Sort vs Quick Sort – Theory

Algorithm	Strategy	Time Complexity	Space Complexity	Stable
Merge Sort	Divide & Conquer	O(n log n)	O(n)	Yes
Quick Sort	Divide & Conquer	Avg: O(n log n), Worst: $O(n^2)$	O(log n)	No

Merge Sort:

- Consistent performance
- Requires extra memory

Quick Sort:

- Faster on small inputs
- Can degrade on sorted or nearly sorted data

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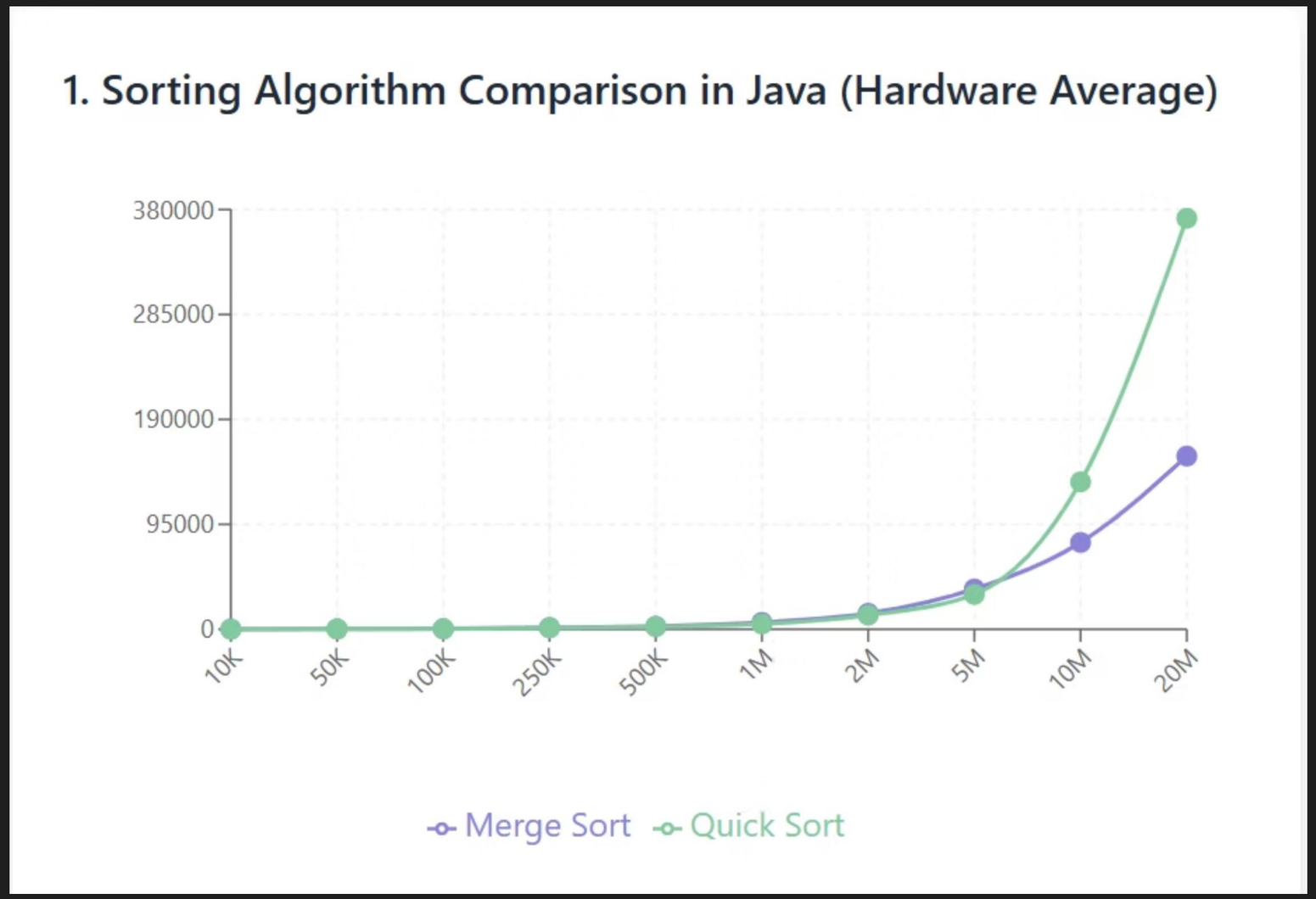
Sorting Algorithm Comparison In Java



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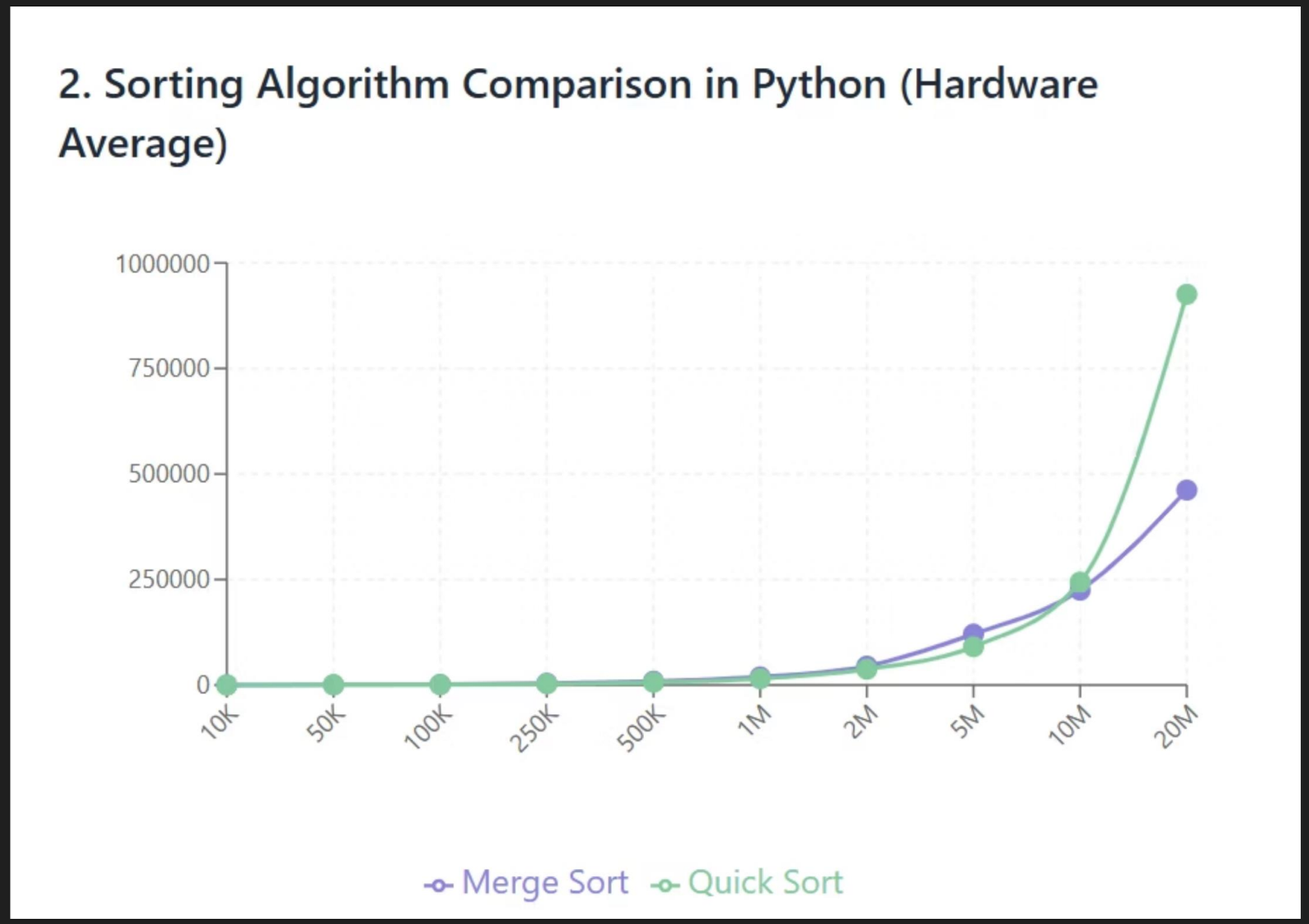
Sorting Algorithm Comparison In Python



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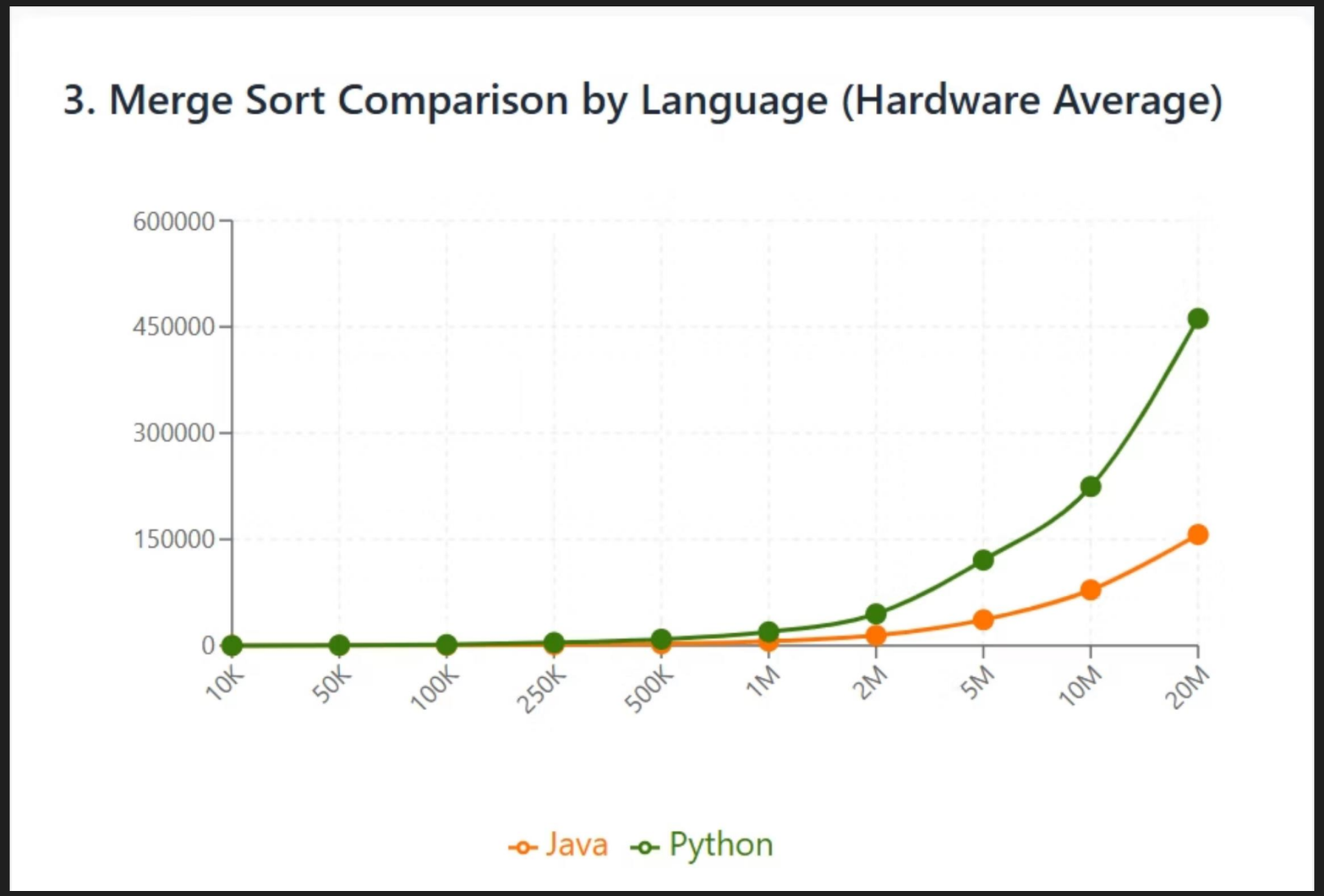
Merge Sort Comparison By Language



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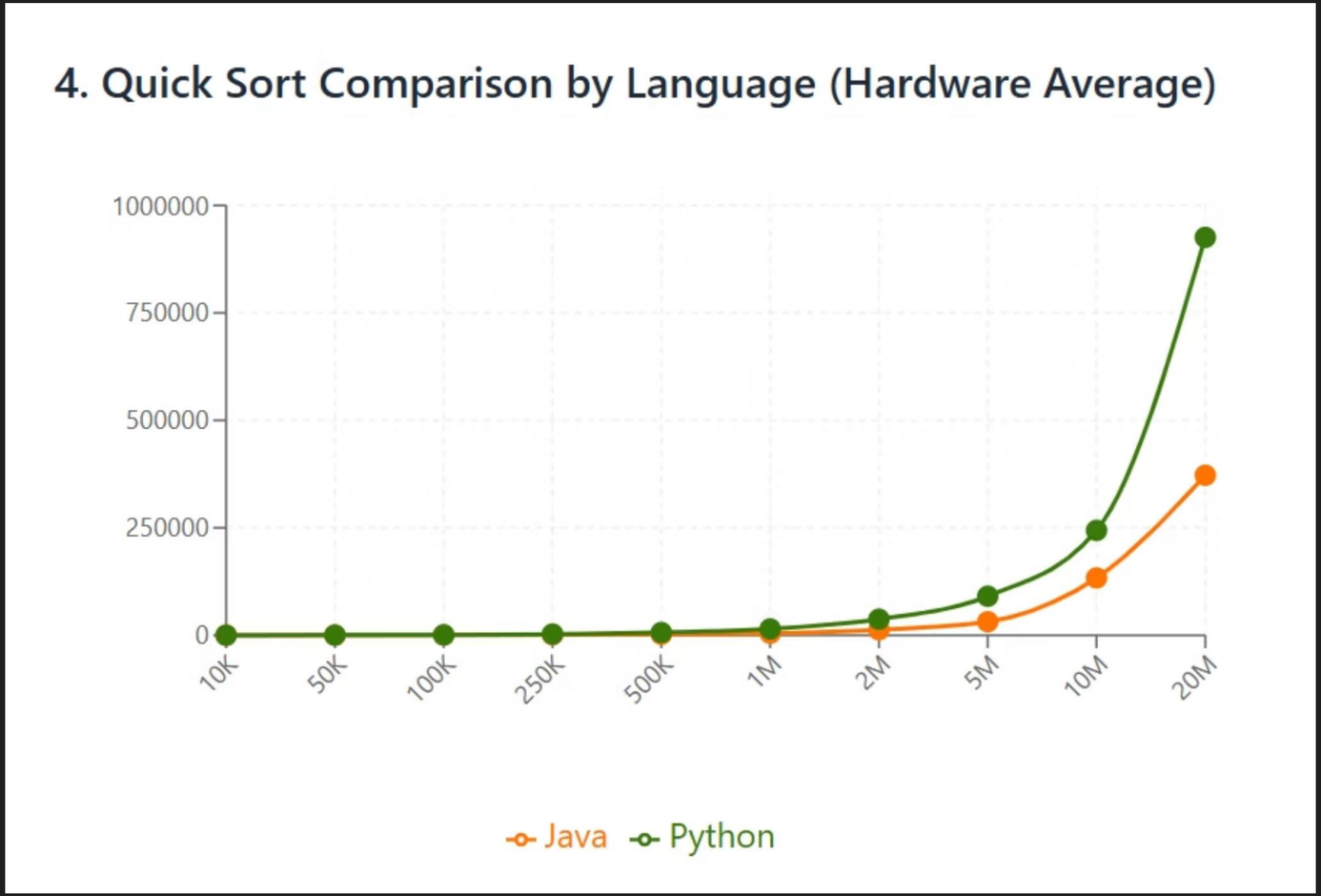
Quick Sort Comparison By Language



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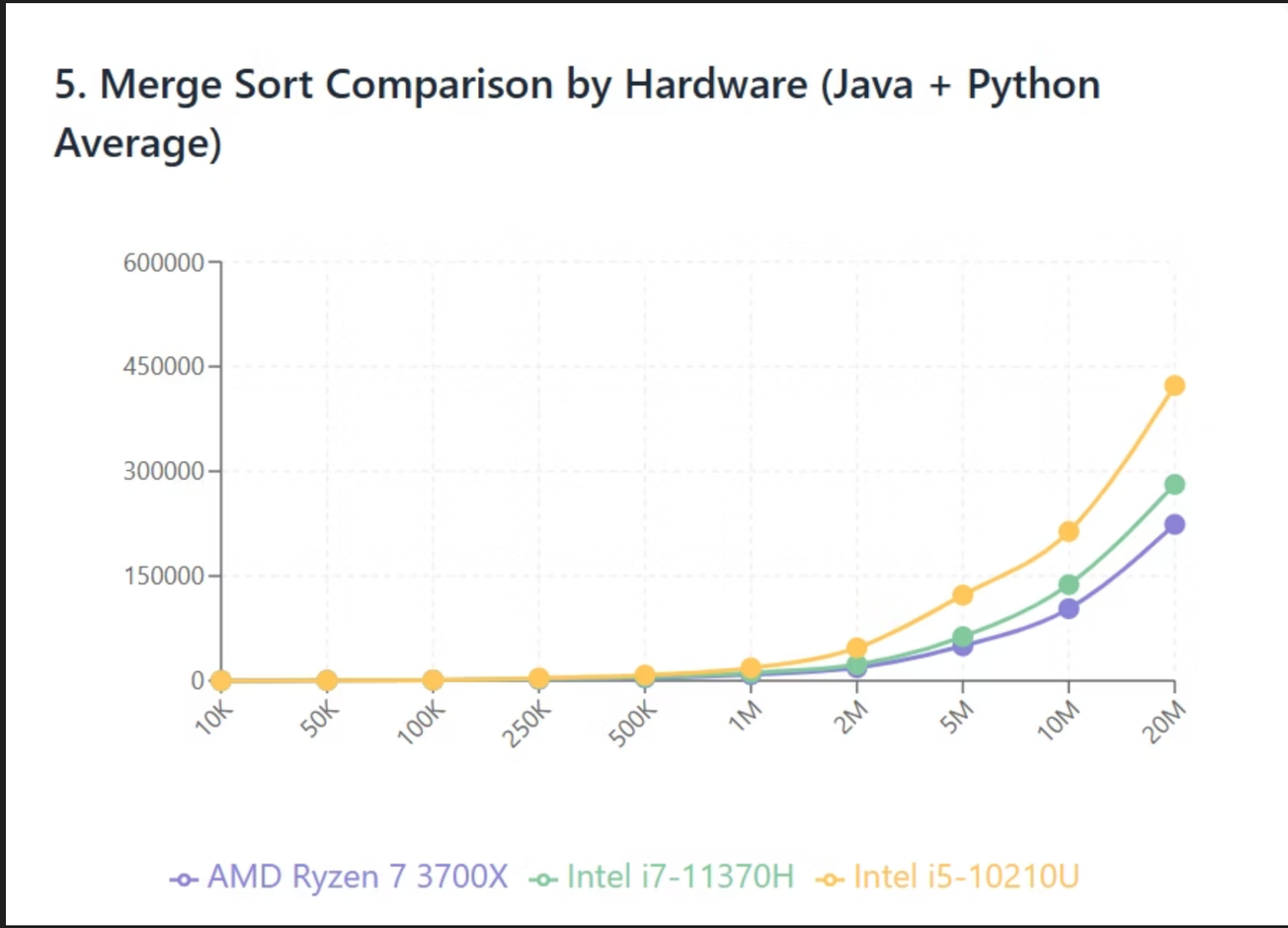
Merge Sort Comparison By Hardware



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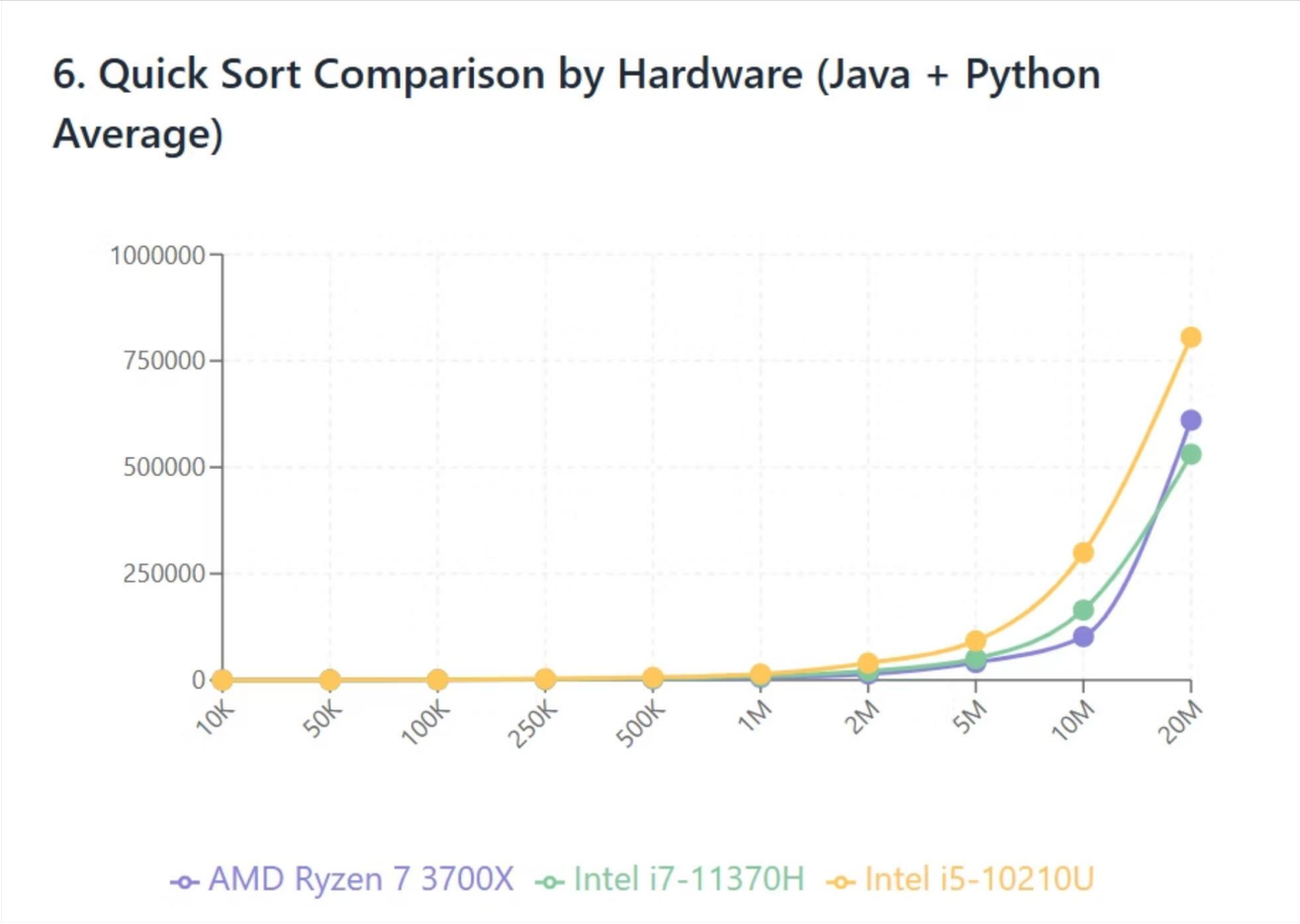
Quick Sort Comparison By Hardware



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Sorting Algorithm Performance

Java Outperforms Python

Speed advantage over Merge Sort and Quick Sort.

Algorithm Efficiency Varies by Input Size

Quick Sort is faster on small datasets, but **Merge Sort** performs better on large inputs.

Hardware Strongly Impacts Performance

Ryzen 7 and i7 CPUs sort large datasets much quicker than i5.

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Best Sorting Algorithms

For large and consistent array sorting, **Merge Sort outperforms Quick Sort** in reliability, scalability, and predictability.

Consistent Speed

Unlike Quick Sort, Merge Sort avoids worst-case slowdowns. It **guarantees O(n log n)** time across all input types.

Predictable Memory Use

Though Merge Sort uses **O(n) space**, its non-recursive structure avoids deep call stacks and is more stable across environments.

Excellent Scalability

Merge Sort handles **large datasets efficiently**, showing **stable performance growth** as input size increases — ideal for heavy workloads.

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Binary Search – Theoretical Analysis

Best Case:

Target found at center → O(1)

Average/Worst Case (Target not found):

O(log n)

Space Complexity:

O(1) for iterative approach

Assumptions:

- Input data must be sorted
- Search based on integer field (index 0 of dataset row)

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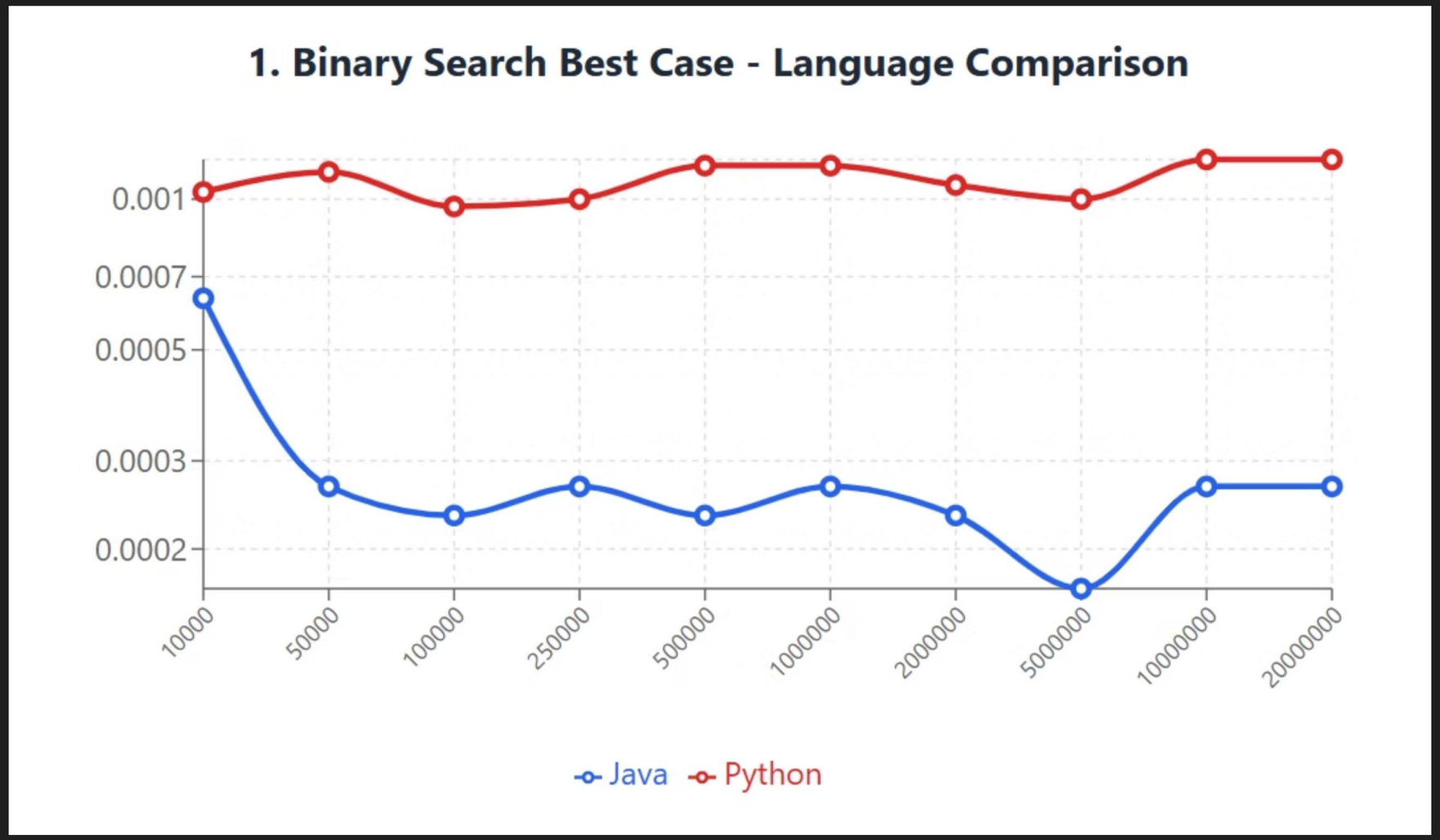
Binary Search Best Case Comparison By Language



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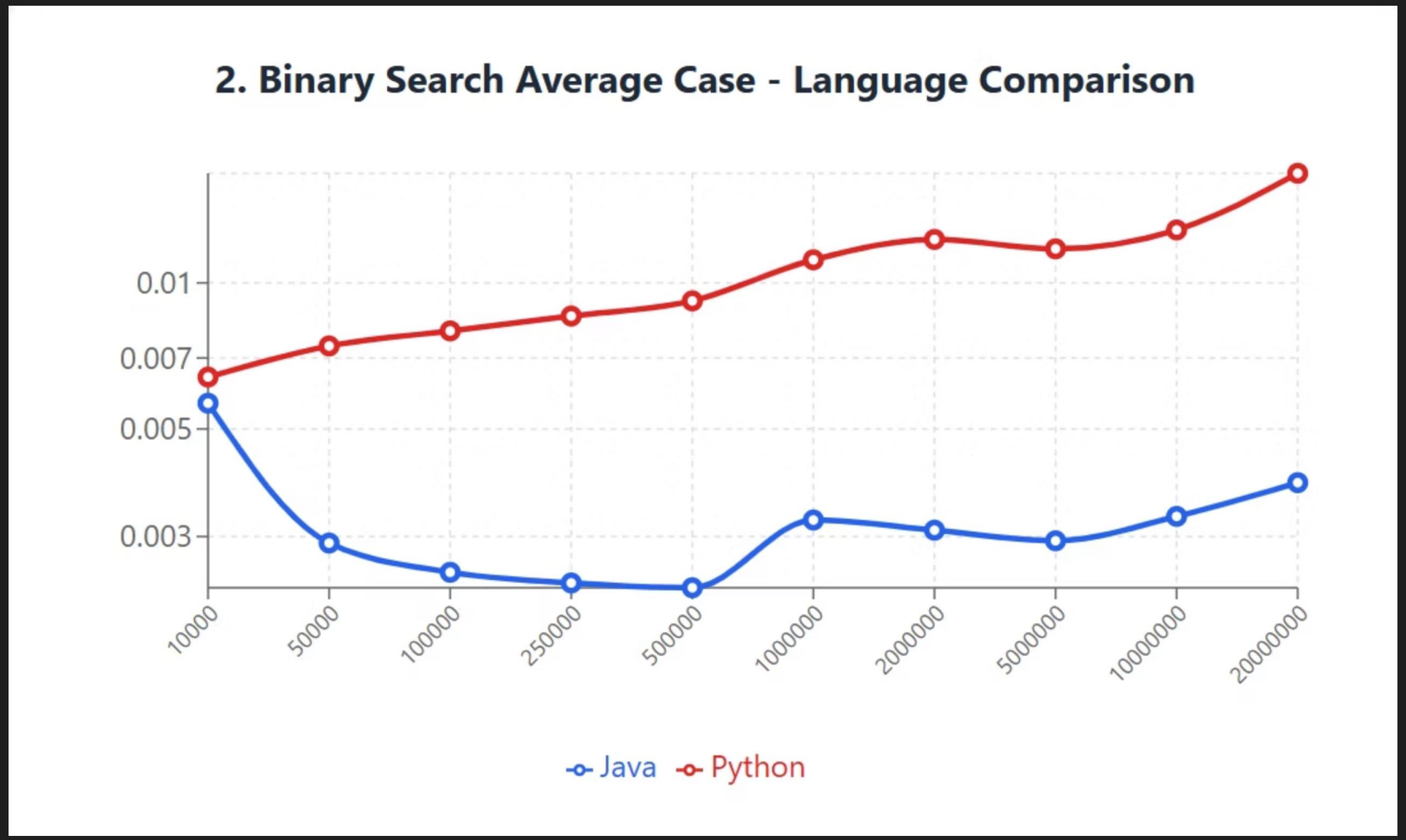
Binary Search Average Case Comparison By Language



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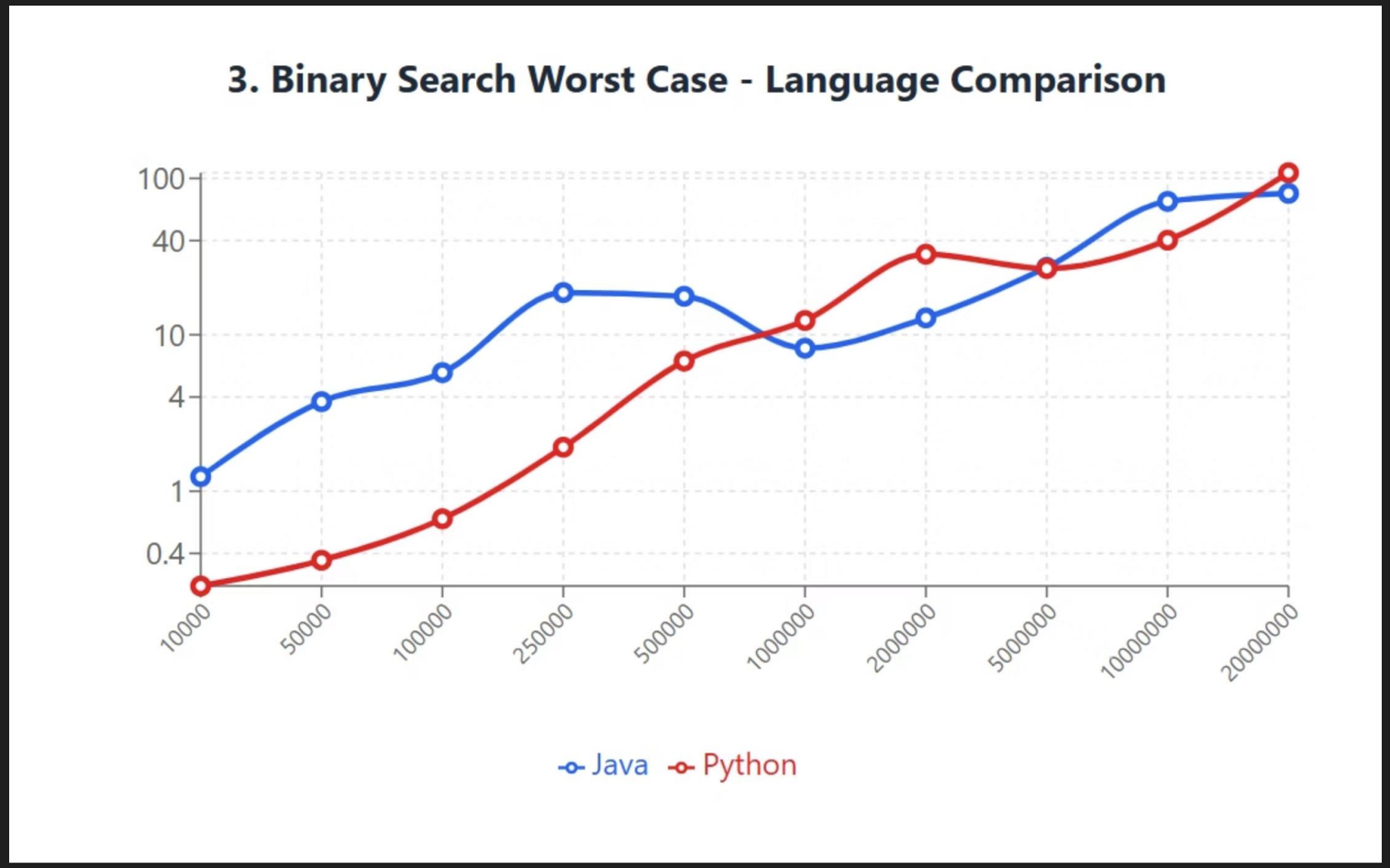
Binary Search Worst Case Comparison By Language



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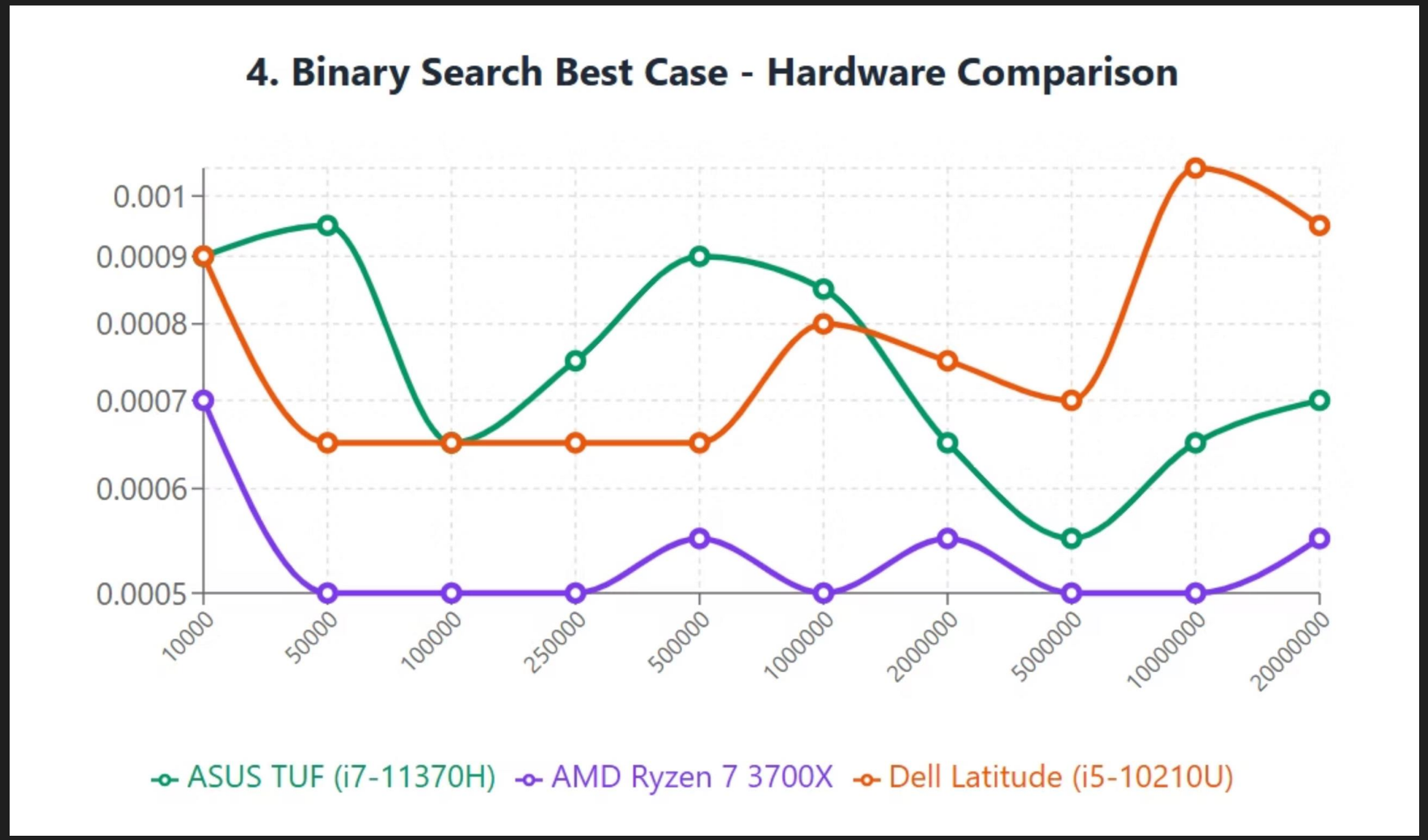
Binary Search Best Case Comparison By Hardware



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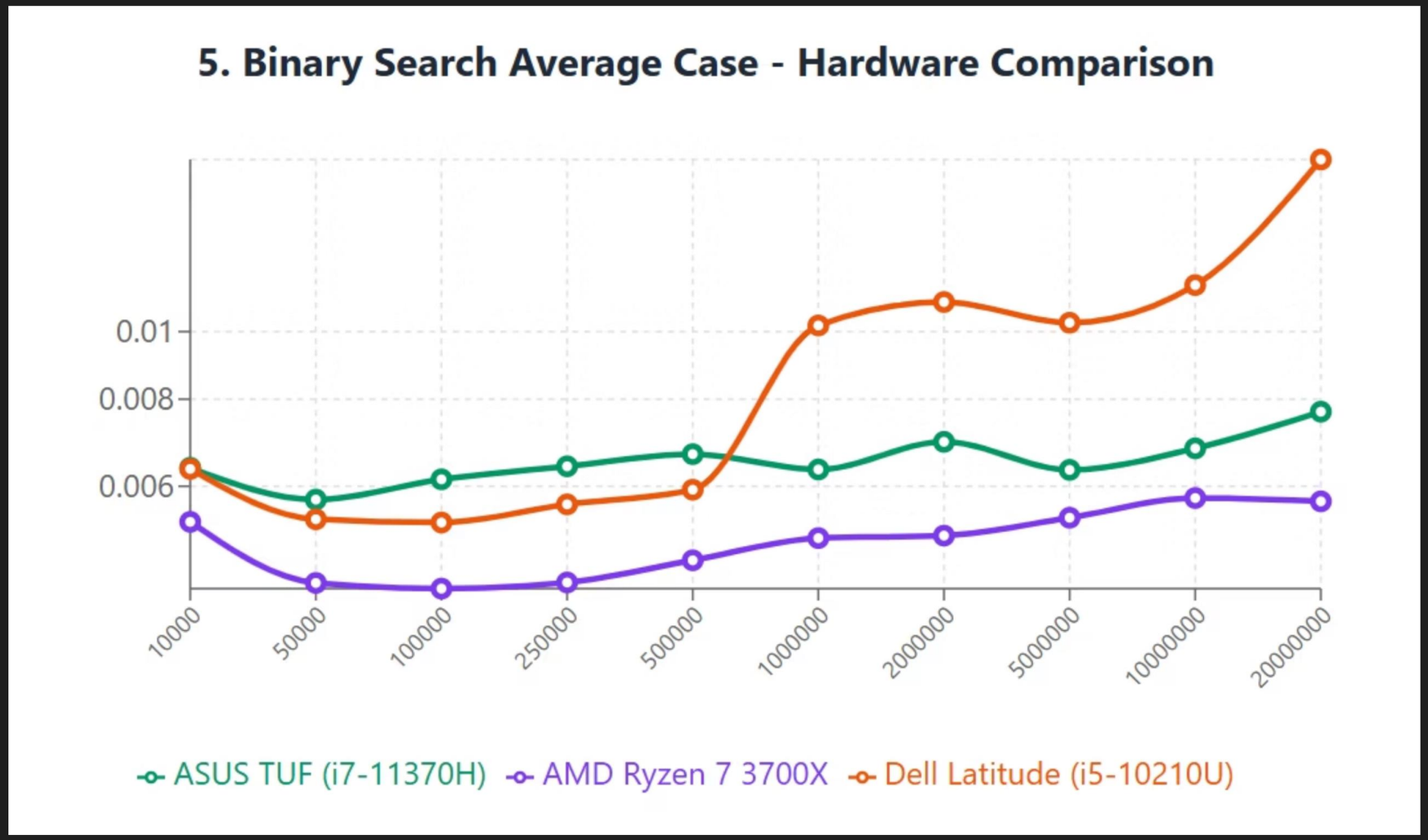
Binary Search Average Case Comparison By Hardware



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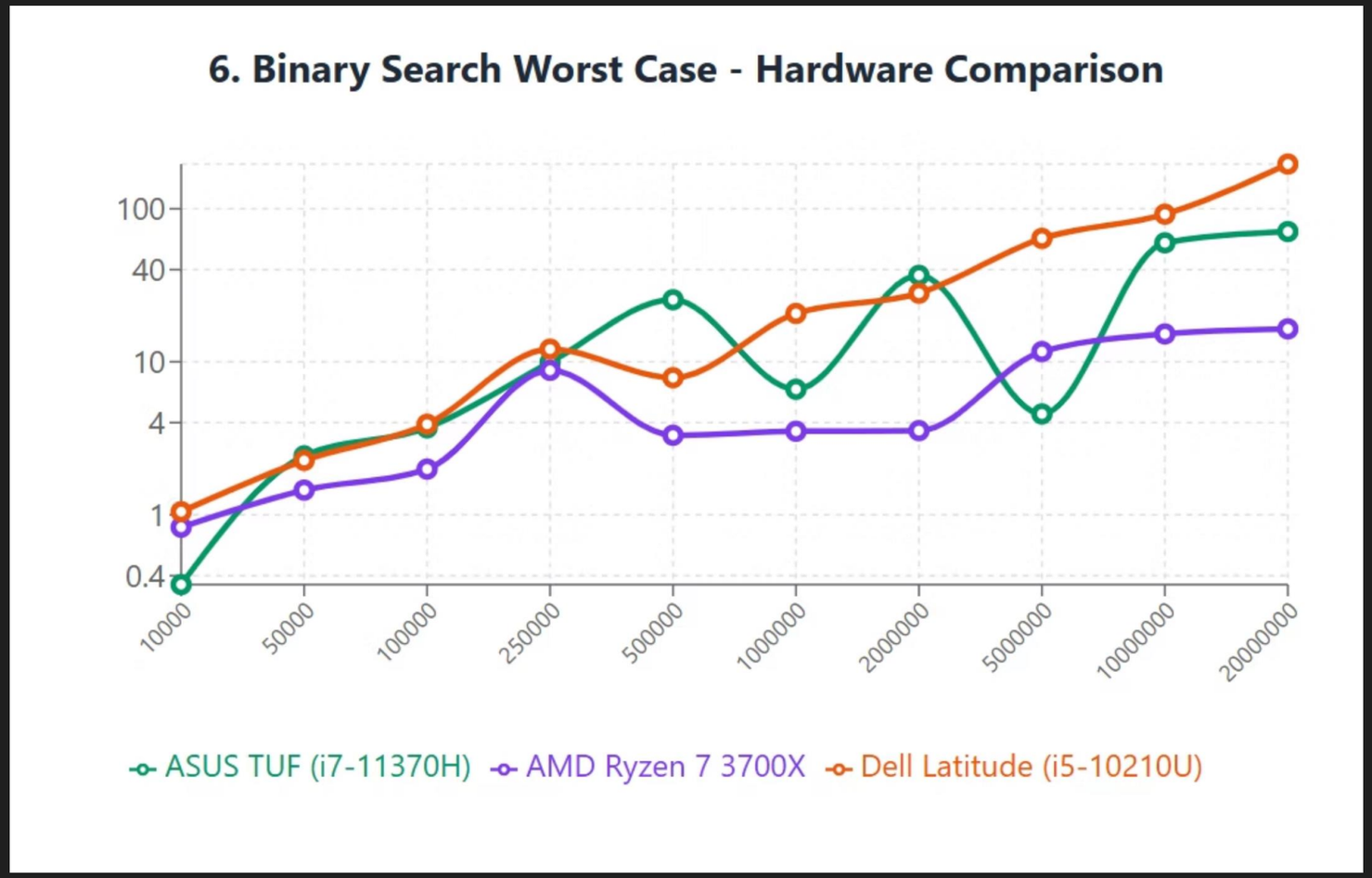
Binary Search Worst Case Comparison By Hardware



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Binary Search Key Insights

Java Outperforms Python in Binary Search Performance

Java consistently shows lower execution times than Python for binary search across all cases (best, average, and worst).

AMD Ryzen 7 3700X Offers Superior Binary Search Performance Among Tested Hardware

The AMD Ryzen 7 3700X consistently demonstrates the fastest binary search times in best, average, and worst-case scenarios compared to the other tested hardware.

Hardware Significantly Impacts Binary Search Performance

Different hardware configurations lead to noticeable variations in binary search execution times, with some hardware performing significantly better than others.

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AVL Implementation Comparison

Criteria	Array-Based AVL	Linked AVL
Insertion	Requires re-sorting	Balanced automatically
Search	Fast (Binary Search)	Fast (Tree traversal)
Space Efficiency	Better	Higher (pointers)
Best Use Case	Static datasets	Dynamic modifications

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References (APA)

Our work draws upon foundational texts and current resources in algorithm analysis and language documentation.

Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2009). **Introduction to algorithms** (3rd ed.). MIT Press.

GeeksforGeeks. (2025). **Merge sort and Quick sort explained**. https://www.geeksforgeeks.org

Python Software Foundation. (2025). **Documentation**. https://docs.python.org

Oracle. (2025). **Java SE Documentation**. https://docs.oracle.com/javase/8/docs/

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Device Specifications

Member	Screenshot/Platform
OW KA SHENG	Dell Latitude 3410, i5-10210U, Win11

Device specifications Copy

Device name	DESKTOP-HFAR9HR
Processor	Intel(R) Core(TM) i5-10210U CPU @ 1.60GHz 2.11 GHz
Installed RAM	8.00 GB (7.77 GB usable)
Device ID	1615ED31-1909-4ACD-8F8B-1C0AAC01F57D
Product ID	00330-53287-21574-AAOEM
System type	64-bit operating system, x64-based processor
Pen and touch	No pen or touch input is available for this display

Related links Domain or workgroup System protection Advanced system settings

LUSUPOV ALIMZHAN	Desktop-HFAR9HR, i5-10210U, 8GB RAM

Device specifications Copy

Device name	DESKTOP-HFAR9HR
Processor	Intel(R) Core(TM) i5-10210U CPU @ 1.60GHz 2.11 GHz
Installed RAM	8.00 GB (7.77 GB usable)
Device ID	1615ED31-1909-4ACD-8F8B-1C0AAC01F57D
Product ID	00330-53287-21574-AAOEM
System type	64-bit operating system, x64-based processor
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Related links Domain or workgroup System protection Advanced system settings

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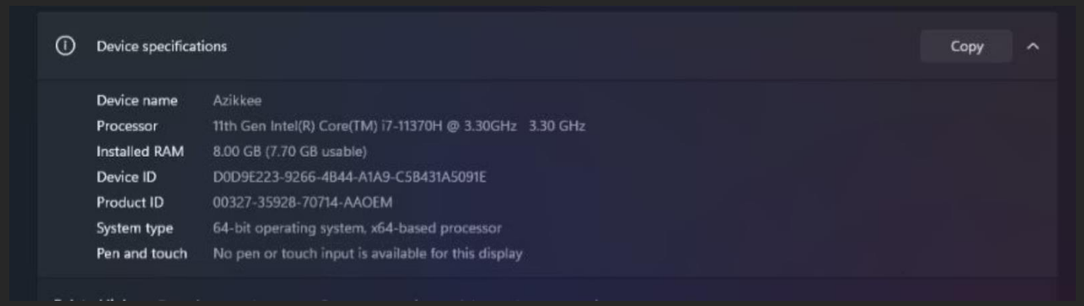
Device Specifications

MUHAMMAD HAZIM BIN MUHD SHAFIQ ZULHILMI KEE

ASUS TUF Dash F15, i7-11370H, Win11



Device specifications

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Device name	Azikkee
Processor	11th Gen Intel(R) Core(TM) i7-11370H @ 3.30GHz 3.30 GHz
Installed RAM	8.00 GB (7.70 GB usable)
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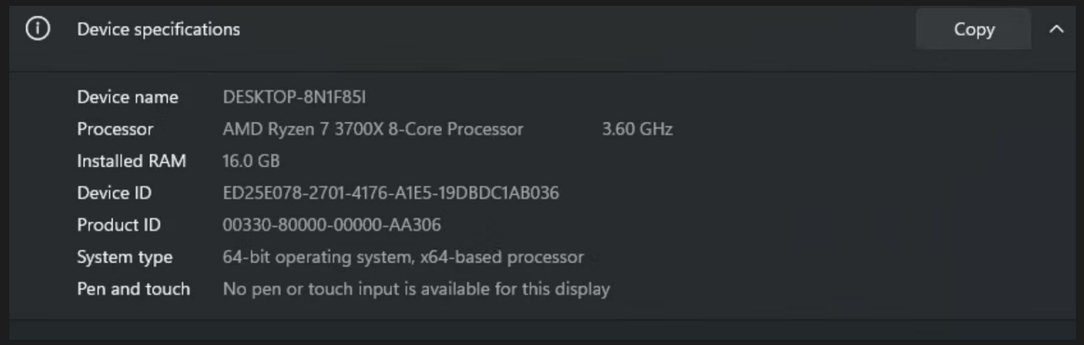
Product ID	00327-35928-70714-AAOEM
System type	64-bit operating system, x64-based processor
Pen and touch	No pen or touch input is available for this display

GALAL OSMAN GALAL MOHAMED

DESKTOP-8N1F85I - AMD Ryzen 7 3700X 8-Core Processor - Windows 11 Pro - 23H2



Device specifications

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Installed RAM	16.0 GB
Device ID	ED25E078-2701-4176-A1E5-19DBDC1AB036
Product ID	00330-80000-00000-AA306
System type	64-bit operating system, x64-based processor
Pen and touch	No pen or touch input is available for this display

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